MILITARY SPECIFICATION

WOOD LAMINATES, SOUTHERN PINE

(FOR SHIP AND BOAT USE)

1. SCOPE

1.1 Scope.- This specification covers laminated Southern pine members for ship and boat use. Such material is suitable for use as straight and curved structural members in small craft, patrol craft, minesweepers, other wooden vessels, and for other applications where structural Southern pine laminates with fully waterproof durable adhesive bonds are desired.

1.2 Classification.- Laminations used shall consist of the following grades, types, densities, and classes:

Grade: (strength quality):
- Grade A
- Grade B
- Grade C
- Grade D

Type: (orientation of annual rings).
- Type 1 - Mixed grain.
- Type 2 - Flat grain.
- Type 3 - Edge grain.

Density:
- Medium grain, or better.
- Dense material.

Class: (Decay susceptibility).
- Class 1 - Decay resistant.
- Class 2 - Low decay resistance.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
- TT-P-320 - Pigment, Aluminum; Powder and Paste, For Paint.
- TT-W-535 - Wood Preservative; Fluor-Chrome-Arsenate-Phenol Mixture.

MILITARY
- MIL-V-1174 - Varnish, Spar, Water Resisting (Formula No. 80).
- MIL-W-6110 - Wood; Determination of Moisture Content of.
- MIL-W-18142 - Wood Preservative Solutions; Oil Soluble; Ship and Boat Use.
- MIL-A-22397 - Adhesive, Phenol and Resorcinal Resin Base (for Marine Service Use).

FSC 5510

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STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and drawings required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 245-64T - Methods for Establishing Structural Grades of Lumber.

(Copies may be obtained from the American Society for Testing and Materials 1916 Race Street, Philadelphia, Pennsylvania 19103.)

SOUTHERN PINE INSPECTION BUREAU

1963 Standard Grading Rules for Southern Pine

(Copies may be obtained from the Southern Pine Inspection Bureau, 520 National Bank of Commerce Building, New Orleans, Louisiana 07150.)

AMERICAN WOOD PRESERVER'S ASSOCIATION (AWPA) PUBLICATIONS

M2 - Standard Instructions for the Inspection of Preservative Treatment of Wood.
P5 - Standard for Water-Borne Preservatives.
A2 - Methods for Analysis of Water-Borne Preservatives and Fire Retardant Formulations.
A3 - Methods for Determining Penetration of Preservatives.

(Copies may be obtained from the Secretary-Treasurer of the American Wood Preserver's Association, 839 17th Street, N.W., Washington, D.C. 20006.)

U.S. DEPARTMENT OF COMMERCE

Commercial Standard CS 253 - Structural Glued Laminated Timber.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the Loading of Forest Products on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 Van Buren Street, Chicago, Ill. 60605.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Prior to award of contract or order, when a manufacturer has not produced laminates within the past 24 months conforming to this specification, evidence shall be submitted certifying the following (see 6.5):

(a) Tests on laminates produced show conformance with the test requirements of this specification as evidenced by an identified report (see 3.6).
(b) The manufacturer is equipped to continue to produce material of quality equal to that tested, in the amount of the order and within the delivery date as evidenced by plant facilities, manpower and quality control capability (see 4.1.1). Appendix A shall be used as a guide to determine the adequacy of the facilities and personnel producing laminates under this specification. Evidence of conformance under Commercial Standard CS 253 may be offered for review in fulfillment of these requirements subject to the detail requirements of this specification.

3.2 Lamination quality.- Laminations used in laminated assemblies shall be one piece for length and width. Edge-joints and end joints may be used to form laminations of the required dimensions. Laminations (includes edge-glued and end-glued components) used in laminated assemblies shall be free from splits, surface checks over 1/32 inch wide or over 4 inches long, ring shake, honeycomb, collapse, case-hardening, or oversteaming (see 4.5.2), wane, and decay in any stage.

3.2.1 Moisture content.- The moisture content at the time of gluing shall be not less than 10 nor more than 15 percent. The moisture content of the core as determined by a distribution section (see figure 1) taken from any board shall not vary from that of the shell by more than 2 percent.

3.2.2 Grade of lumber.- Grades and maximum allowable defects shall be in accordance with table I, 3.2.2.1 and 3.2.2.2 (see 6.2). Allowable knot and equivalent defect sizes for the strength ratios specified in table I shall be the permissible size for the given strength ratio. As specified in table II and table IV of ASTM D245-64T (strength ratios for joists and planks for knots on faces and edges within middle third of length). Knot sizes shall apply to full length of faces and edges (see 2.2). Slope of grain and knot sizes shall be measured in accordance with 4.5.3 and 4.5.4 respectively. For laminates intended for remanufacture into smaller cross-sections, defect limitations shall apply to end use dimensions.

3.2.2.1 Slope of grain.- When measured as specified in 4.5.3, slope of grain along the length of any face or edge of a lamination shall not be steeper than the maximum slope specified in table I for the applicable grade.

3.2.2.2 Knots and equivalent knot defects.- The sums of diameters of knots and equivalent knot defects for various lamination grades shall be as specified in table I. Knot and equivalent knot defect limitations are based on the finished lamination width within the finished dimension of the Southern pine laminate in which it appears.

3.2.2.2.1 Knots.- All knots shall be sound. Knot clusters are not permissible. Intergrown knots containing checks will be equivalent to a sound tight knot of the same size. All knots appearing on edge surfaces, and on face surfaces of laminations less than 3 inches wide shall be tight. In laminations 3 inches in width and wider, knots on the face surface larger than 3/8 inch in diameter, shall be tight. The diameter of knots shall be determined as specified in 4.5.4.

3.2.2.2.2 Burls.- Burls shall be considered as equivalent knot defects of the same size.

3.2.2.2.3 Holes.- Holes are not permissible on edge surfaces, and face surfaces less than 3 inches wide. In laminations 3 inches and wider, knot holes and holes due to mechanical damage averaging 3/8 inch and less in diameter appearing on face surface shall be considered the same as a sound tight knot of the same size. Holes greater than 3/8 inch in diameter are not permissible in any location.

3.2.2.2.4 Pitch pockets and bark pockets.- Equivalent knot defect values for pitch pockets and bark pockets are specified in table II. Pitch pockets with soft or exuding pitch are not permissible.

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Table I - Permissible slope of grain, knots, and equivalent knot defect limitations, for lamination grades.

Lamination grade	Maximum slope of cross grain expressed as one inch rise in inches of length	Maximum sum of diameters of knots and equivalent knot defects, expressed as percentage strength ratio (see 3.2.2).
Grade AA	1 in 18	85
Grade A	1 in 15	72
Grade B	1 in 12	65
Grade C	1 in 10	55
Grade D	1 in 8	50

Note (1) Defect limitations apply to finished laminate width. Defect size for 7 inch width (actual) shall be maximum permitted regardless of width over 7 inches.

(2) The following lumber grades of Southern pine (SPIB rules 1963) have been recommended by the Southern Pine Inspection Bureau as being essentially equivalent to the specified lamination grades. However, because of variations between this specification and SPIB rules, laminations must be re-graded according to this specification prior to utilization.

Lamination	Is equivalent to
Grade AA	Industrial 86 KD Southern pine
Grade A	Industrial 72 KD Southern pine
Grade B	Industrial 65 KD Southern pine
Grade C	Industrial 58 KD Southern pine
Grade D	Industrial 50 KD Southern pine

Table II - Equivalent knot defect values for pitch pockets and bark pockets.

Pitch pocket (or bark pocket)	Equivalent knot defect	
	Flat-grain laminations Inch	Edge-grain lamination Inch
1/16 inch in width and 3 inches in length, or 1/8 inch wide and 2 inches in length	1/4	1/8
1/8 inch in width and 4 inches in length, or 1/4 wide and 2 inches in length	3/8	1/4

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Table II - Equivalent knot defect values for pitch pockets and bark pockets (cont'd.).

Pitch pocket (or bark pocket)	Equivalent knot defect	
	Flat-grain laminations Inch	Edge-grain lamination Inch
1/8 inch in width and 8 inches in length, or 1/4 inch in width and 4 inches in length, or 1/4 inch wide and 2 inches in length	1/2	3/8
3/16 inch in width and 6 inches in length, or 3/8 inch in width and 4 inches in length	5/8	1/2

3.2.3 Type (orientation of annual rings) (see 6.2 and 6.7).-

3.2.3.1 Type 1 (mixed grain).- Laminates may contain either flat grain or edge grain, or both and individual lamination components may contain both flat and edge-grain.

3.2.3.2 Type 2 (flat grain).- The annual growth rings in laminations and lamination components shall be so oriented that tangent lines to the annual growth rings form an angle of less than 45 degrees with the wide surface of the piece.

3.2.3.3 Type 3 (edge-grain).- The annual growth rings in laminations and lamination components shall be so oriented that tangent lines to the annual rings form an angle of more than 45 degrees with the wide surface of the piece.

3.2.4 Density (see 6.2).-

3.2.4.1 Medium grain lumber.- Averages on either one end or the other not less than 4 annual rings per inch, measured over a 3 inch portion of a radial line from pitch as in dense, or over such portion less than 3 inches as may be present, and which shall be representative of average growth on cross section.

3.2.4.2 Dense lumber.- Averages on either one end or the other not less than 6 annual rings per inch and not less than 1/3 summerwood measured as specified in Standard Grading Rules for Southern Pine Lumber. Contrast in color between summerwood and spring wood must be distinct. Pieces averaging less than 6 annual rings per inch and not less than 4 meet dense requirements if averaging 1/2 or more summerwood.

3.2.5 Class (see 6.2).-

3.2.5.1 Class 1 (decay resistant).- Laminations shall be free from sapwood, or shall be treated with preservative in accordance with 3.2.5.1.1 through 3.2.5.1.4.

3.2.5.1.1 Preparation for treatment.- The wood before treatment shall be sufficiently seasoned to a moisture content not in excess of 25 percent in the sapwood for proper impregnation with preservative.

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3.2.5.1.2 Preservative material (fluorchrome-arsenate-phenol mixture).- The preservative chemicals shall conform to either TT-W-535, type A or B. The treating solution shall be of a concentration suitable to produce a retention of 0.50 lbs/ft.^{3} of dry salt in the sapwood portion.

3.2.5.1.3 Treatment.- The treatment shall be any method which does not use a pressure higher than 150 pounds per square inch (p.s.i.) or a temperature higher than 140° Fahrenheit (F.), and provides 100 percent penetration of preservative in the sapwood with a dry salt retention of not less than 0.50 lbs/ft.^{3} of sapwood.

3.2.5.1.4 Conditioning of wood after treatment.- Lumber shall be dried to the required moisture content by either air drying or kiln drying, using temperatures less than 150°F.

3.2.5.2 Class 2, low decay resistance.- The area of edge or wide surfaces of any lamination may contain up to 20 percent unstained sapwood.

3.3 Adhesives.- Adhesives used for all gluing operations shall conform to MIL-A-22397. Only products which have been approved for listing on the applicable Qualified Products List shall be utilized.

3.4 Construction.- A laminate shall consist of layers of laminations joined together by the adhesive specified in 3.3 in accordance with the process requirements of 3.5.

3.4.1 Lamination thickness.- All laminations in a given laminate shall be approximately the same thickness. The maximum lamination thickness for straight laminates shall be 1-3/4 inches. The maximum permissible lamination thickness for curved laminates shall be computed from the relationship:

$$\frac{\text{Thickness}}{\text{Radius of curvature}} = \frac{1}{110} = .009$$

When bending laminations to curved forms, breakage of any lamination, failure of any end-joint, or observance of compression wrinkles, splitting, splintering or other indications of bending damage, shall result in rejection of the laminate unless the damaged lamination is replaced by a sound one within the assembly time of the adhesive used.

3.4.2 Arrangement of end joints.- Laminated assemblies may contain end joints in individual laminations of the design specified in 3.5.1.2. End joint spacing within the laminate shall meet one of the following arrangements as specified (see 6.9 and 6.2).

Arrangement 1 - Spacing of adjacent end joint tips at a glueline shall be not less than 6 inches. Any plane at right angles to the longitudinal axis of the laminate that intersects an end joint shall not intersect any part of an end joint in the adjacent lamination to either side. Any 6 inch long section of the finished laminate shall not contain more than 1/4 of the total number of laminations in the 6 inch section.

Arrangement 2 - Within the outer 1/4 of the total depth of the laminate spacing of adjacent end joint tips at a glueline shall be not less than 6 inches. Within the center 1/2 of the depth of the laminate, the end joint spacing shall be not less than 2 inches. Any plane at right angles to the longitudinal axis of the laminate that intersects an end joint shall not intersect any part of an end joint in the adjacent lamination to either side.

3.4.3 Finished dimensions.- Laminated Southern pine members shall be manufactured to full dimensions required by the procuring activity and shall conform to the pattern lines as determined from templates or drawings furnished (see 6.2).

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3.5 Process requirements.- Edge-gluing of lumber components to produce laminations of required width and end-gluing of lumber components to form laminations to required length shall procede final surfacing and gluing of laminations. For laminations wider than 18 inches, one edge joint may be glued at the time of laminating. All other edge joints shall be pre-glued. All edge joints shall meet the requirements of 3.6.3.2.

3.5.1 Machining of surfaces to be bonded.-

3.5.1.1 Edge-surfaces.- Strips of Southern pine used to build up laminations for width by edge-gluing shall have a minimum width of 1-1/2 inches at the time of fabrication. Strips to be edge glued shall be machined to produce a uniform flat (without tongue and groove, or similar construction) and true surface with a minimum of damage to wood fibers.

3.5.1.2 End-surfaces.- Laminations built up for length by end-joining of lumber components shall require use of a plain flat scarf-joint with a slope not steeper than one inch rise in 12 inches of length.

3.5.1.3 Surfacing laminations.- Laminations shall be dressed to a uniform thickness throughout, and the total thickness variation of any lamination at the time of gluing shall not exceed a maximum of 0.01 inch, as determined through the use of an accurate thickness gage. Surfaces shall not show manufacturing defects to an extent which will prevent sufficient contact between gluing surfaces for complete adhesion. Surfaces shall not be sanded to remove such defects after surfacing.

3.5.2 Adhesive spread.- The adhesive shall be uniformly applied to both contact faces.

3.5.3 Assembly.- At the time of gluing the temperature of lumber materials and ambient air temperature shall be 75° ± 15°F., and surfaces to be bonded shall be free of oil, dirt, crayon marks, and other foreign material which may interfere with bonding of the adhesive.

3.5.4 Clamping.- Application of pressure to glue lines shall be accomplished within the assembly period limitations required for the particular adhesive used. Cauls, clamps, and clamp spacing shall be adequate so as to uniformly apply and hold a glue line pressure of 125 p.s.i. with a plus tolerance of 50 p.s.i. throughout the cure. During initial application of pressure, clamps shall be loose enough to allow slippage of material into proper position. During application of final pressure, clamping procedure shall be such as to allow glued surfaces to be forced into intimate contact at all points.

3.5.4.1 Edge-joints.- Bearing surfaces on clamps, and clamp spacing shall be such as to apply and hold required pressure without damage to material.

3.5.4.2 End-joints.- Clamping procedure shall not cause splitting of feathered edges or other material damage which will affect the strength.

3.5.4.3 Laminate assemblies.- Either male or female forms for gluing bent members are acceptable. Laminate assemblies shall be provided with top and bottom cauls of oak or equally hard and strong materials of such thickness, which in conjunction with suitable clamps and clamp spacing, will provide the required glue line pressure uniformly, and which will not result in damage to the laminate material. Clamps with equalizing pressure heads or equally effective pressing equipment shall be uniformly spaced close enough to develop pressure as specified at all points. Spacing of the clamps or units will vary with the number and thickness of the cauls used and the pressure required. In straight or curved members, clamping shall begin at any one point and proceed progressively toward the ends.

3.5.5 Curing.- In the curing process, a glue line temperature of not less than 120°F. shall be maintained for the time period specified by the adhesive manufacturer for the particular approved adhesive, except that the glue line temperature shall not be less than 120°F. for straight laminates and 150°F. for curved laminates. The chamber temperature shall not exceed 215°F. for untreated lumber and 170°F. for 60 minutes for preservative treated lumber. The curing temperature shall be determined at the innermost glue line through the use of an accurate potentiometer and thermocouples, while assemblies are under full

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clamp pressure. Where assemblies are large, or where more than one assembly is cured in the same curing chamber a sufficient number of determinations shall be made to assure that glue line temperatures are being met throughout the curing chamber.

3.5.5.1 Edge-joints.- During the curing operation, temperature and relative humidity controls shall be maintained in such a manner that all applicable requirements of 3.2 are met throughout the lamination length.

3.5.5.2 End-joints.- During the curing operation temperature and relative humidity controls shall be maintained in such a manner that all applicable requirements of 3.2 are met throughout the lamination length.

3.5.5.3 Laminate assemblies.- A dry kiln or equally effective curing chamber shall be provided with adequate controls for uniformly maintaining the temperature and humidity conditions required in the curing process. If clamped assemblies are to be moved into a separate curing chamber for the curing process, they shall not be removed from the laminating jig, and the portable jig shall be rigid enough to prevent shifting of the assembly in the jig. Clamped assemblies shall be placed in the curing chamber before the temperature in the chamber reaches 100°F. The curing period shall commence when the innermost glue line temperature reaches the temperature required for the particular approved adhesive, and shall continue for the required curing period. The laminate shall then be cooled under humidity conditions controlled to prevent surface checking.

3.6 Test requirements.-

3.6.1 Moisture content of finished laminate.- The finished material shall have a moisture content not less than 10 percent nor more than 15 percent (see 4.5.1).

3.6.2 Block shear (see 4.5.5).- The average block shear strength between laminations shall be not less than 1100 p.s.i. with at least 75 percent wood failure. No individual glue line shall have a shear strength less than 750 p.s.i., unless additional block shear tests conducted on that glue line prove the deficiency to be local.

3.6.3 Resistance to delamination.- The maximum permissible delaminations shall be as specified in 3.6.3.1, 3.6.3.2 and 3.6.3.3 (see 4.5.6).

3.6.3.1 Between laminations.- Delamination between laminations shall either not exceed 5 percent at the end of the second cycle or 8 percent at the end of the third cycle.

3.6.3.2 Edge-joints.- Where a single edge-joint appears in a test specimen, delamination shall either not exceed 10 percent at the end of the second cycle, or 20 percent at the end of the third cycle. Where two or more edge-joints appear in a specimen, average delamination shall either not exceed 5 percent at the end of the second cycle, or 10 percent at the end of the third cycle.

3.6.3.3 End-joints.- Delamination in end-joints shall either not exceed 5 percent at the end of the second cycle, or 10 percent at the end of the third cycle. Delamination limits apply to individual end-joints.

3.7 Finished laminate.- Finished laminates shall show no signs of opening up at the glue line, or damaged fibers from improper use of cauls and clamps.

3.8 Coatings.-

3.8.1 Preservative treatment.- After machining, laminated assemblies shall be coated on ends and sides with one coat of preservative solution conforming to type A or B of MIL-W-18142. The preservative shall be allowed to dry before applying paint.

3.8.2 End coating.- The ends of laminated Southern pine shall be painted with two coats of aluminum or mica paint after the preservative has dried for at least 24 hours. Aluminum paint shall consist of 2

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pounds of aluminum paste, conforming to type II, class B of TT-P-320, and 1 gallon of phenolic varnish, conforming to MIL-V-1174. A drying interval of 24 hours shall elapse before applying a second coat. An alternate end coating may be any suitable commercial end coating which is satisfactory to the bureau or agency concerned.

3.8.3 Side coating.- When specified (see 6.2) and 6.6) and after preservative required by 3.8.1 has dried, the sides of the laminated Southern pine shall be painted with two coats of aluminum paint as in 3.8.2.

3.9 Marking.- When specified (see 6.2), each member shall be marked to identify the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Supplier quality control system.- The supplier shall provide and maintain a quality control system acceptable to the Government for the supplies covered by this specification.

4.1.2 Inspection verification.- All quality control operations performed by the contractor shall be subject to verification at unscheduled intervals. Verification shall consist of, (a) surveillance of the operations to determine whether the practices, methods, and procedures are being properly applied, and (b) product inspection to measure quality of product offered for acceptance. Instances of poor practices which might have an affect upon the quality of the product shall be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to the prescribed acceptance criteria of this specification has been demonstrated. To minimize interference with operations, the contractor will designate a responsible official or officials to whom the inspector will report such instances, and who will alert the inspector of action taken to correct production difficulties which occur. Repeated failures to meet specification requirements will result in cancellation of the contract or order.

4.2 Inspection lot.-

4.2.1 Lamination components.- All components or pieces prior to end-joining shall be considered a lot.

4.2.2 Laminations.- A lot shall consist of all laminations submitted for inspection at the same time.

4.2.3 Laminate.- A lot shall consist of all laminates submitted for inspection at the same time, which are manufactured for approximately the same end use using a single lot of adhesive, and are fabricated under approximately the same temperature and process conditions as determined by process inspection.

4.3 Sampling for examination and tests.- For the sampling specified in 4.3.1 through 4.3.4 the "single sampling plan" of MIL-STD-105 is applicable.

4.3.1 Sampling for examination and test of laminations (or components).- For the examinations and tests specified in 4.4.1, laminations (or components prior to end-joining where specified) shall be sampled on an unbiased random basis in accordance with table III and MIL-STD-105. Where applicable the same sample may be utilized in inspecting for more than one defect.

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Table III - Sampling levels for examination and test for various lamination defects, corresponding acceptable quality levels, and basis for acceptance or rejection.

Requirement	Inspection level	Acceptable Quality Level for lot acceptance (percent)	Basis for rejection of lot
Grain, dense (see 3.2.4.2) and table I)	I-components prior to end-joining	4.0 medium grain	Any board is poorer than medium grain, or if the number of boards with medium grain exceed the acceptance number
Grain, medium (see 3.2.4.1)	I-components prior to end-joining	None	a. Individual boards have less than 4 rings per inch.
Orientation of annual rings (flat or edge-grain) (see 3.2.3)	I-components prior to end-joining	4.0 grain not as specified	a. Where either grain is specified, any lamination is predominantly the wrong grain. b. Where either grain is specified the number of laminations with components which do not have the required grain exceeds the acceptance number
Casehardening (see 3.2)	S-3-lamination components prior to end-joining (test 4.5.2)	4.0-casehardened or over steamed	a. The number of laminations which are casehardened or oversteamed exceeds the quality acceptance number. (Individual laminations which have components which are casehardened or oversteamed shall be rejected.)
The combined AQL for all of the following counted together shall be 15 percent			
Class (see 3.2.5)	I (board with sapwood where preservative is used)	4.0 for untreated wood 1.5 for preservative treated wood	a. The number of laminations which do not meet the required class requirement, exceeds the acceptance number. (Individual laminations which do not meet the class requirement shall be rejected.)
Moisture content	I (test 4.5.1)	6.5-for moisutre content up to two percent outside limitations	a. Any lamination has components which are more than 2 percent outside required limitations. b. Number of laminations which have components with moisture up to two percent outside moisture content laminations exceeds the quality acceptance number.

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Table III - Sampling levels for examination and test for various lamination defects, corresponding acceptable quality levels, and basis for acceptance or rejection (cont'd.).

Requirement	Inspection level	Acceptable Quality Level for lot acceptance (percent)	Basis for rejection of lot
Excess cross grain (see 3.2.2.1)	I	4.0-within next lower grade	a. Any lamination contains cross grain which falls below the next lower grade (slope of 1 in 6 for grade D). b. The number of laminations which have cross grain falling within the next lower grade, exceeds the acceptance number.
Knot defects	I	4.0-next lower grade or exceeding 1-1/2 inch in diameter.	a. Laminations fall below next lower grade. b. The number of laminations which have knot defects which fall within the next lower grade or, in excess of 1-1/2 inches, exceeds the acceptance number.
Surface checks	I	4.0	a. The number of laminations which do not meet the requirements of 3.2 exceed the acceptance number.
Edge-joints	I examination only	6.5-wood strips too narrow	a. Any lamination shows a defective edge-joint glue line. b. The number of laminations which show strips too narrow exceeds the acceptance number.
End-joints	I-examination only	None	a. Any lamination contains end-joints with too steep slope, or visually defective bonds, or contains wood at the end-joint which is charred, split, or otherwise damaged to the extent which will cause loss in strength.
Other non-permissible lamination defects	I	None	Any lamination contains one or more of the following or any other defect which affects strength, durability, or glueability. (1) Soft or incipient decay in any amount. (2) Ring shake. (3) Soft, or oozing pitch. (4) Honeycomb, or collapse. (5) Splits. (6) Oils, crayon marks, wax, etc. on surfaces to be bonded.

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4.3.1.1 Sampling for test of end-joints.- When a laminator has not produced wood laminates with end-joints conforming to this specification for the previous 12 months, seven end-joints shall be fabricated in accordance with regular production procedure, and all seven shall be prepared and tested as specified in 4.4.1.1. When a laminator has produced wood laminates in the previous 12 months, whose end-joints met specification requirements, each production week throughout the period of production, an end-joint shall be selected from a regular production lamination for test in cyclic delamination as specified in 4.4.1.1.

4.3.2 Sampling for use of suitable adhesive, adhesive use, process and facilities.- A systematic and effective quality control procedure shall be maintained by the laminator to provide an effective and accurate examination as outlined in 4.4.2.

4.3.3 Sampling for examination of finished laminates.- Sampling for the examination specified in 4.4.3 shall be in accordance with inspection level II of MIL-STD-105. The Acceptable Quality Level for minor defects such as dimension slightly off, and slight material degrade due to curing or mechanical damage during manufacture shall be 4.0 percent defective.

4.3.4 Sampling for test of finished laminates.- Sampling for test of finished laminates shall be in accordance with 4.3.4.1, 4.3.4.2 or 4.3.4.3 as applicable.

4.3.4.1 No previous production experience.- When a laminator who has not previously produced marine laminates conforming to this specification is beginning production, specimens for test shall be selected from each member, until five consecutively produced members have passed all requirements of this specification. Thereafter, samples shall be selected for test in accordance with inspection level III of MIL-STD-105 until fifteen consecutively tested, major members, such as frames and stems, have met the requirements. Thereafter, members shall be sampled in accordance with 4.3.4.2.

4.3.4.2 Continued production.- When a laminator has met the prerequisites in 4.3.4.1, specimens shall be selected for moisture content, block shear and cyclic delamination tests on an unbiased random basis in accordance with tables IV and V (column headed "Normal sampling") and MIL-STD-105. The acceptable quality level for moisture content shall be AQL = 4.0 percent defective. For laminates with shear strength not more than 50 p.s.i. below the required minimum, or wood failures percent of 70 to 75 percent, the acceptable quality level shall be as follows:

AQL = 1.5 percent defective when sample is less than lot size.
AQL = 4.0 percent defective when sample is the entire lot.

Sampling shall shift from "normal" to "reduced" or "tightened" sampling when the criteria specified under these column headings in tables IV and V are met.

4.3.4.3 Lapse in production.- When a laminator has not produced laminates conforming to this specification for a period of 12 months, specimens for test shall be selected from each member, until five consecutively produced members have passed all requirements of this specification. Thereafter samples shall be selected for test in accordance with 4.3.4.2.

4.4 Lot acceptance examination and test.-

4.4.1 Examination and test of laminations.- Laminations (or components) selected in accordance with 4.3.1 shall be inspected to determine conformance with the applicable requirement. Acceptance or rejection of lamination (or component) lots shall be in accordance with table III. Where edge-joints are used to form lamination for width, edge-joint mating surfaces shall be examined for slope of grain requirement prior to edge-gluing.

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Table IV - Sampling for moisture content and block shear tests.[1/]

Member	Sampling level		
	Reduced sampling	Normal sampling	Increased sampling
	Moisture content shear strength and percent wood failure requirements consistently met in all specimens tested in 5 preceding consecutive lots).		One or more test specimens in 3 preceding consecutive lots failed to meet moisture content of at least one part of the shear requirement; or if the lot is sampled for retest.
All members directly exposed to salt water and/or severe exterior exposure - (stems, keels, skegs, horn timbers, garboards, bilge keels, masts, etc.)	S-3	S-4	III
All interior members not directly exposed to salt water (except filler blocks)	S-2	S-3	II
Filler blocks	S-1	S-2	I

[1/] Moisture content and block shear are separate requirements to be considered independently when shifting to "reduced" or "increased" sampling.

Table V - Sampling for cyclic delamination tests.

Member	Sampling level		
	Reduced sampling	Normal sampling	Increased sampling
	Percent delamination requirement met in all specimens tested in 3 preceding consecutive lots an average value for all specimens below 5 percent delamination		At least one test specimen in 3 immediately preceding consecutive lots failed to meet percent delamination requirement

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Table V - Sampling for cyclic delamination tests (cont'd.).

	Sampling level		
	Reduced sampling	Normal sampling	Increased sampling
All members directly exposed to salt water (stems, keels, skegs, horn timbers, garboards, bilge keels, etc.)	S-3	S-4	III
All interior members not directly exposed to salt water (except filler blocks)	S-1	S-2	I
Interior filler blocks)	None	None	None

4.4.1.1 Test of end-joints.- When an end-joint sample is selected for test as specified in 4.3.1.1 a fifteen inch section of lamination shall be cut so that the centerline of the end-joint is at the center of the section. The section shall be glued as the center lamination of a test beam with four other laminations of the same width, thickness, and length. A specimen 3 inches long, and the full cross section of the block shall be prepared so that the center of the end-joint is at the center of the specimen. The specimen shall be tested as specified in 4.5.6. Failure of 2 consecutive end-joints to meet the requirement of 3.6.3.3 shall be basis for rejection of material containing end-joints until test results are presented which shows that the laminator can on a production basis fabricate at least seven consecutive end-joints meeting the requirements of 3.6.3.3.

4.4.1.2 Examination and test for preservative (class I only).- A 0.2 inch diameter boring shall be taken from each sample board with sapwood selected in accordance with table III, so that the boring extends through the deepest section of the sapwood into the heartwood. The borings shall be examined and tested in accordance with 4.5.7.1, for preservative penetration. One-half of the sample borings shall be analyzed for retention (4.5.7.2). If desired by the laminator, full cross-sections may be substituted for borings for the examinations and tests specified in 4.5.7.1 and 4.5.7.2.

4.4.2 Use of suitable adhesive, adhesive use, process and facilities.- In accordance with the effective procedure specified in 4.3.2, inspection shall be made to determine use of suitable adhesive, meeting adhesive use requirements, process requirements, and proper calibration, performance, and use of facilities. Failure to use suitable adhesive will result in rejection of any lot involved. Failure to conform to adhesive use requirements, process requirements, and proper calibration performance and use of facilities will result in immediate temporary rejection of lots until the difficulty has been diagnosed and corrected. Acceptability of involved laminates shall be based on results of the inspector's diagnosis as to seriousness. Failure to take effective corrective action shall result in rejection of subsequent lots. Repeated failure to adhere to the general laminating requirements set forth shall require that production be stopped until corrective action is effected.

4.4.3 Examination of finished laminates.- Each of the sample laminates selected in accordance with 4.3.3 shall be examined to determine conformance with the requirements of the specification, in so far as can be determined by visual and dimensional inspection. If any laminate in the sample does not conform to requirements for lamination thickness, end-joint spacing, deviates considerably from required dimensions or shape, or has visually defective adhesive bonds, it shall be rejected, pending individual examination of

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each laminate. All laminates with one or more of these major defects shall be rejected. (Members which show defective adhesive bonds shall be rejected even though specimens taken from the ends of that member passed the requirements of 3.6.) Also, if the total number of all minor defects in the sample exceed the acceptance level listed in 4.3.3, the lot represented by the sample shall be rejected.

4.4.4 Test of finished laminates.-

4.4.4.1 Moisture content.- Moisture content of finished laminates shall be determined in accordance with 4.5.1. If the number of laminates not meeting the moisture content requirements of 3.6.1 exceed the acceptance level listed in 4.3.4, the lot represented by the sample shall be rejected.

4.4.4.2 Block shear test of interlamination glue line.- Each of the samples selected for block shear tests in accordance with 4.3.4 shall be tested in accordance with 4.5.5. Failure of any sample member to conform to the requirements of 4.3.4 shall result in rejection of the lot, and every member of the lot shall be tested individually. The rejection of two lots in succession for failure to conform to 3.6.2 shall require that the manufacturer develop test information which shows that the difficulty has been diagnosed and corrected, the requirements can be met, and the quality control system has been corrected to effectively screen out defective laminates. The rejection of 3 lots in succession is considered sufficient reason for stopping production until corrective action is taken.

4.4.4.3 Cyclic delamination test.- Samples selected for cyclic delamination tests in accordance with 4.3.4 shall be tested in accordance with 4.5.6. Failure of any of the sample members to conform to all of the requirements of 3.6.3.1 and 3.6.3.2 shall be basis for rejection of the lot involved, and every laminate in the lot shall be tested individually. All individual laminates not meeting the requirements of 3.6.3.1 and 3.6.3.2 shall be rejected. The rejection of any members in 2 lots in succession for failure to conform with 3.6.3.1 shall require that the manufacturer present test information which indicates that the difficulty has been diagnosed and corrected, and the requirements can be met. Failure of 2 consecutive edge-joint samples to meet the requirements of 3.6.3.2 shall be basis for rejection of material containing edge-joints until the manufacturer presents test information which indicates that requirements can be met. Consecutive failures of interlamination glue line in excess of those listed above shall be considered sufficient reason for stopping production until corrective action is taken.

4.5 Test methods.-

4.5.1 Moisture content.- Moisture content in lumber in laminations shall be determined using an accurate moisture meter or oven test in accordance with MIL-W-6110. If a moisture meter is used, it shall be capable of determining moisture content at the shell and core. Determination of moisture content of treated laminations may be by any method yielding results complying with 3.2.1. Moisture content of finished laminates shall be by oven test.

4.5.2 Casehardening.- Casehardening shall be determined in accordance with figure 1D.

4.5.3 Slope of grain measurement.- Slope of grain shall be measured on faces and edges of laminations. Where edge-joints or end-joints are used, slope of grain on mating surfaces shall be measured prior to assembly. Slope of grain measured on a given face or edge is considered as the general slope of grain along that length of face or edge. This disregards normal local variations around permissible knots, knot holes, and other acceptable imperfections. This also disregards other local slope of grain variations which may have been caused by such defects as knots which originally occurred in the trimmed-off portion of the board, provided these areas are not more than 8 inches in length, are not more than twice as steep as the permissible general slope of grain, and that these areas are not spaced closer than 4 feet along the length of the piece.

4.5.4 Measurement of knots.- Knot sizes shall be measured according to the method depicted in figure 2 "Measurement of Knots in Joists and Planks", of ASTM D 245-64T.

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4.5.5 Block shear test.-

4.5.5.1 Test specimens.- Adhesive lines in a given sample shall be clearly marked with ink in arabic numerals. Adhesive lines shall then be sampled for test on an unbiased random basis in accordance with sampling level II of MIL-STD-105, and test specimens prepared. The remainder of the material shall be retained for verification testing of other glue lines if necessary. The test specimens shall be cut as shown on figure 2 and may conform to either (A) or (B) so that the grain direction is parallel to the direction of loading during test. Care shall be taken in preparing the test specimen to make the loading surfaces smooth and parallel to each other and perpendicular to the height. When sawing the bonded assembly, care shall be exercised to insure that the saw cuts extend to but not beyond, the adhesive line. The width and height of the specimen at the adhesive line shall be measured to the nearest 0.010 inch to determine the shear area.

4.5.5.2 Apparatus.- The testing machine shall be fitted with a compression shearing tool containing a self-aligning seat to insure uniform lateral distribution of the load. The shearing tool shown on figure 3 has been found satisfactory. For range of loads encountered in test, indicated loads shall be accurate within ± 2.5 percent of true values as determined by standard procedures for verification of testing machine. Increments of load on the indicating device shall be in easily readable divisions between one and 2.5 percent of the value of loads encountered when test specimens break. The load indicating device shall include damping apparatus which permit reading of ultimate loads under conditions of sudden failure. Provisions should be included for a fairly uniform rate of loading to failure. The load shall be applied with a continuous motion of the movable head to a maximum load at a rate of 0.025 inch per minute with a permissible variation of ± 25 percent.

4.5.5.3 Procedure.- The test specimen shall be placed in the shearing tool so that the load may be applied as specified in 4.5.5.2. The loading shall continue until failure. The shear stress at failure shall be calculated in pounds per square inch, by dividing load at failure by the area of the adhesive line area between the two laminations measured to the nearest 0.01 square inch. Also the percent wood failure for each adhesive line shall be estimated to the nearest five percent. The average percent wood failure and shear strength shall be calculated for each specimen. If either the average percent wood failure or shear strength fail to meet the requirements of 3.6.2, the remainder of the adhesive lines in the test specimen shall be prepared and tested, and a new average for shear strength and percent wood failure shall be computed which includes all adhesive lines in the test specimen.

4.5.6 Cyclical delamination test.-

4.5.6.1 Test specimen.- Specimens for cyclical exposure test shall be 3 inches in length in the direction of the grain and full cross section of the member. When an end-joint appears in the cyclical exposure test specimen, it shall be located in the sample so that it appears equally on both exposure faces.

4.5.6.2 Test procedure.-

4.5.6.2.1 Vacuum pressure cycle.- The specimens specified in 4.5.6.1 shall be placed in an autoclave, weighted down, and water at a temperature of 65° to 80°F. shall be admitted into the autoclave in quantity sufficient to submerge the specimens completely. Thereafter, a vacuum of 20 to 25 inches of mercury shall be drawn and held for 2 hours. The vacuum shall then be broken, and a pressure of 80 + 5 p.s.i. shall be applied to the submerged specimens and held for 2 hours. The pressure shall then be released and a vacuum of 20 to 25 inches of mercury shall again be drawn and held for 2 hours while the specimens remain submerged. The vacuum shall be released again and a pressure of 80 + 5 p.s.i. applied to the submerged specimens and held for 2 hours. (Elapsed time 8 hours, 40 minutes, including load, unload and place in dryer.)

4.5.6.2.2 Drying period.- The specimens shall be dried for a period of 88 hours (3-2 3 days) in air at 80 to 85°F. and 25 to 30 percent relative humidity, and circulating at the rate of 500 ± 50 feet per minute. During the drying the specimens shall be placed at least 2 inches apart and with the end surfaces parallel to the stream of air.

4.5.6.2.3 Duration of test.- The soaking and drying periods specified in 4.5.6.2.1 and 4.5.6.2.2 shall be repeated once, for a total of 2 cycles of setting and drying, at the end of which period delamination

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shall be measured in accordance with 4.5.6.2.4. If the second cycle test requirements of 3.6.3 are met, the test shall be discontinued. However, if any of the second cycle requirements of 3.6.3 are not met, the soaking and drying cycle shall be repeated once more for a total of 3 cycles of soaking and drying, comprising a total test period of 12 days, and delamination again measured.

4.5.6.2.4 Measurement of delamination.- At the end of the soaking and drying cycle prescribed in 4.5.6.2.3, the total length of each type of glue line (interlamination, edge-joint, scarf-joint) shall be measured to the nearest 0.05 inch on both end grain surfaces of the exposure test specimen. Also, the total length of delamination in each type of glue line shall be measured on both end-grain surfaces. (Delamination is a term used to express separation of wood surfaces at the glue lines. When the separation is in the wood, even though very close to the glue line, it is termed wood failure or checking. Magnification is often necessary to determine whether the failure is in the glue or wood. A feeler gage, 0.004 inch in thickness is convenient for probing into the glue line to determine if separation actually exists.)

4.5.6.2.5 Calculations.-

a. Delamination interlamination glue lines: Percent delamination in interlamination glue lines shall be calculated as follows:

$$\text{Percent delamination} = \frac{\text{Total length of delamination in interlamination glue lines measured on the end grain surfaces of test specimen}}{\text{Total length of interlamination glue lines measured on the end grain surfaces of test specimen}} \times 100$$

b. Delamination in edge-joints: Percent delamination in edge-joints shall be calculated as follows:

$$\text{Percent delamination} = \frac{\text{Total length of delamination in edge-joint glue lines measured on the end grain surfaces of test specimen}}{\text{Total length of edge-joint glue lines measured on end grain surfaces of test specimen}} \times 100$$

c. Delamination in scarf-joints: Percent delamination in scarf-joints shall be calculated as follows:

$$\text{Percent delamination} = \frac{\text{Total length of delamination in scarf-joint glue lines measured on the end grain surfaces of test specimens}}{\text{Total length of scarf-joint glue lines measured on the end grain surfaces of test specimens}} \times 100$$

4.5.7 Determination of preservative penetration and retention (class 1).-

4.5.7.1 Preservative penetration.- Each boring or cross-section shall be examined for extent of preservative penetration in the sapwood. Where doubt exists as to penetration, chemicals in accordance with AWPA A3, method 2 may be used as an aid.

4.5.7.2 Preservative retention.- The entire sapwood of each boring or cross-section shall be separated from the heartwood. The sapwood portions shall be composited, and the composite analyzed by an accurate and recognized assay method.

4.6 Additional inspection and tests.- Where other specifications form a part of this specification, unless otherwise specified herein, sampling, examination and tests shall be conducted as required in referenced specifications.

4.7 Inspection for preparation for delivery.- Laminates which are shipped shall be inspected to verify conformance to the requirements of section 5 herein.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early use.-

5.1.1 Wood laminates.-

5.1.1.1 Packaging.- Laminated frame members shall be bundled in pairs or other suitable grouping, and each member within the bundle shall be suitably marked (see 3.9). Smaller laminated members should be bundled, when practicable. Bundles shall be secured by means of steel strapping. No screws, bolts, or spikes shall be inserted into the members for packaging and packing purposes. All members of the bundle shall be protected against cutting of the strapping by means of strips of plywood, metal, fiberboard or other protective material placed between the bundle members and the strapping. The packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early use.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by carrier and will afford protection against physical damage during handling and direct shipment from the supply source to the using activity for early use. The wood laminates shall be shipped in open or closed cars according to size of the individual member. When loaded in open cars (gondola and flat car) loading shall be in strict accordance with Section 5 of the Association of American Railroads "Rules Governing the Loading of Forest Products on Open Top Cars". The wood laminates, protected as specified (see 3.8) shall be covered with waterproof paper or other suitable weather resistant material. When loaded in closed freight cars or covered vans material shall be loaded in accordance with the methods recommended in Pamphlet No. 20 of the Association of American Railroads or other carrier regulations applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided in accordance with the contractors commercial practice. The information shall include nomenclature, grade, type, density and class, contract or order number, contractor's name and address.

6. NOTES

6.1 Intended use.- Southern pine laminates covered by this specification are suitable for marine use and for other severe exterior exposure, except exposure to marine organisms. The adhesive bonds have high strength, have extreme resistance to salt water and shrinkage and swelling, and have long time durability.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Pre-award certification (see 3.1 and 6.5).
(c) Thickness, width, length, shape (include drawings and sharpest radius of curvature, if curved members are required), and intended use (for example frame, keel, etc.) of laminates (see 3.4.3 and 3.9).
(d) Grade, type, density, and class of laminations (see 3.2 and 6.4 through 6.7).
(e) Arrangement of end-joints in laminates (see 3.4.2 and 6.9).
(f) Coatings required (see 3.8 and 6.6).
(g) That the contractor should pile material awaiting shipment on stickers, and protect it from precipitation and the direct rays of the sun, when directed by the inspector.

6.3 Definitions.-

6.3.1 Laminated wood.- For purposes of this specification laminated wood is considered to be an adhesive-bonded assembly consisting of layers of wood with their grain running in the same direction.

6.3.2 Lamination.- A lamination is a single layer of wood used in a laminated assembly.

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6.3.3 Equivalent knot defects.- Lamination defects listed in this specification which have been assigned a value equivalent to a knot of a given diameter size (see 3.2.2.2).

6.4 Grade of laminations.- The lamination grades developed for this specification are such that they should be readily available. However, the lower quality material is more readily available and less costly than the higher grades. Where permissible by strength requirements, weight limitations, and other design factors, the lower grades should be used where feasible. Southern pine boards are generally manufactured and stocked in fixed nominal widths. In order to reduce material waste and cost, where feasible, lamination widths in the finished laminate form should be based on these readily available nominal board dimensions.

6.5 Pre-award examination.- Pre-award evidence submitted by prospective suppliers shall be forwarded for technical review and recommendations to:

Commander
Quality Assurance Division
Puget Sound Naval Shipyard
Bremerton, Washington 98314

6.6 Side coatings.- Side coatings should be specified if the laminated Douglas fir is stored outdoors, unprotected. End coatings should be renewed as necessary when laminated Douglas fir is stored outdoors in covered piles. When specifying side coatings, the end use should be taken into account. Surfaces in the exterior underwater portions of ships and boats are coated with antifouling paint. Since aluminum coating has a tendency to blister when used under such paints, it should not be specified for these surfaces if the end product is to be supplied ready to install. Provisions for covered transportation and storage should be specified if side coatings are not specified.

6.7 Laminate type.-

(a) Mixed grain.- A general purpose, economical type, for use where repeated wetting and drying will not occur. Cyclic wetting tends to cause excessive distortion due to mixed grain.
(b) Flat grain.- A general purpose, readily available grade, suitable for most applications.
(c) Vertical grain.- A more expensive, special type for use where wearing surface is important or moisture movement across width (perpendicular to annual rings) should be a minimum. This type may be more efficiently produced by vertically laminating flat grain stock.

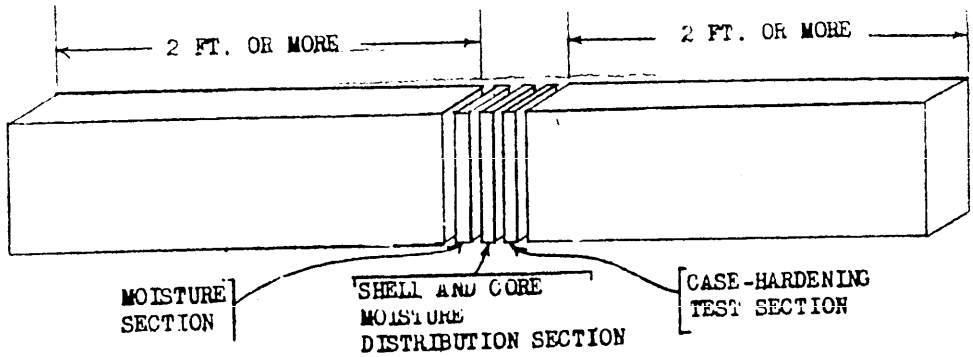
6.8 Laminate class.- For service where deterioration by decay fungi and wood destroying insects is expected, class 1 laminates should be specified. For direct exposure to deterioration by marine borers (Toredo, Bankia, Limmoria), class 2 laminates should be specified to be preservative treated for marine borer resistance after shaping and drilling.[1/]

6.9 End-joint arrangements.- Arrangement 1 is intended for laminates which will be loaded primarily in tension. Arrangement 2 is intended for laminates which will be loaded primarily in bending.

Preparing activity:
Navy - SH
(Project 5510-N014Sh)

[1/] Federal Specification TT-W-571 is recommended for marine borer preservative treatment requirements.

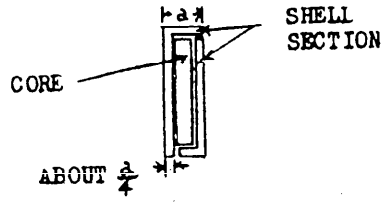
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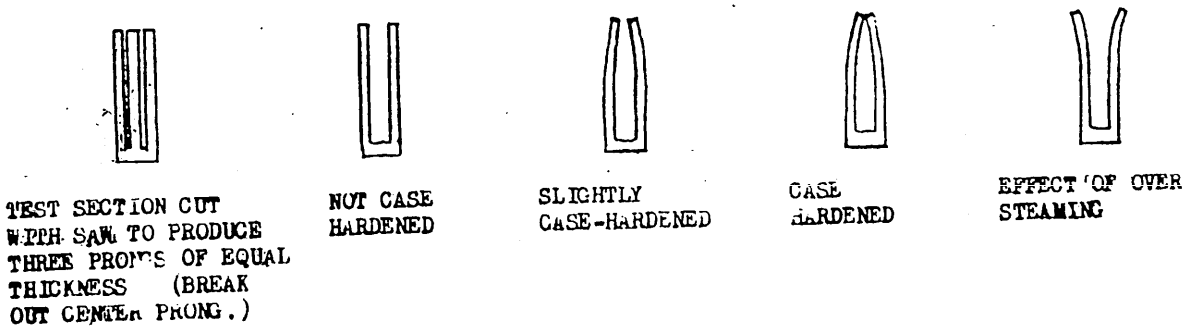
-A- SELECTION OF TEST SECTIONS FOR DETERMINING MOISTURE CONTENT, MOISTURE DISTRIBUTION, AND CASE-HARDENING.



-B- MOISTURE SECTION



-C- SHELL AND CORE MOISTURE DISTRIBUTION SECTIONS



-D- TESTING FOR CASE-HARDENING.

Figure 1 - Test Sections for Determining Moisture Content, Moisture Distribution, and Case-Hardening.

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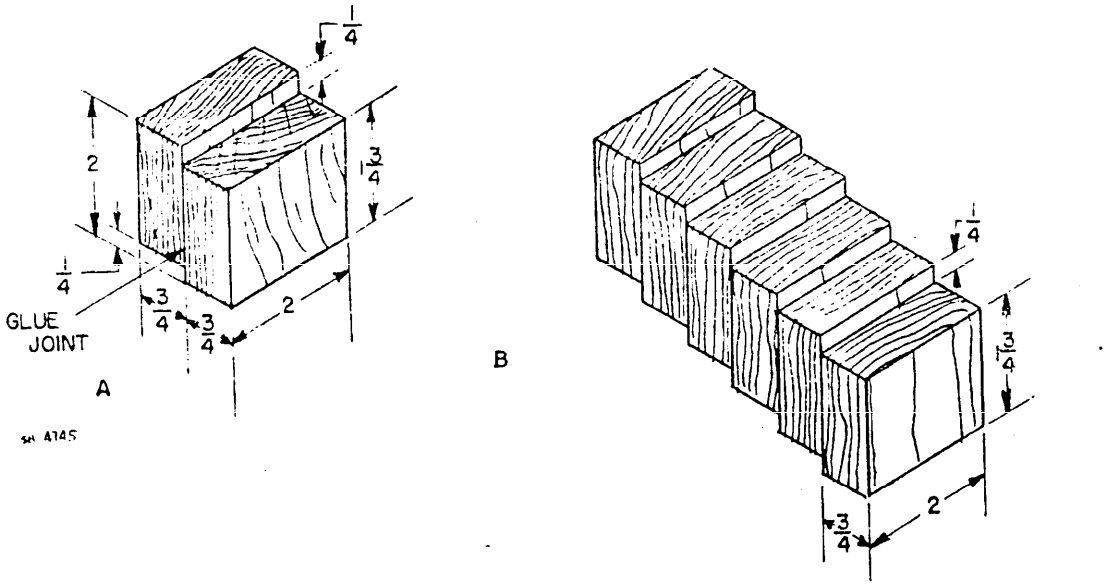


Figure 2 - Standard Block (A) and Stair Step Type (B) Shear Specimens for Testing Glue Joint Strength.

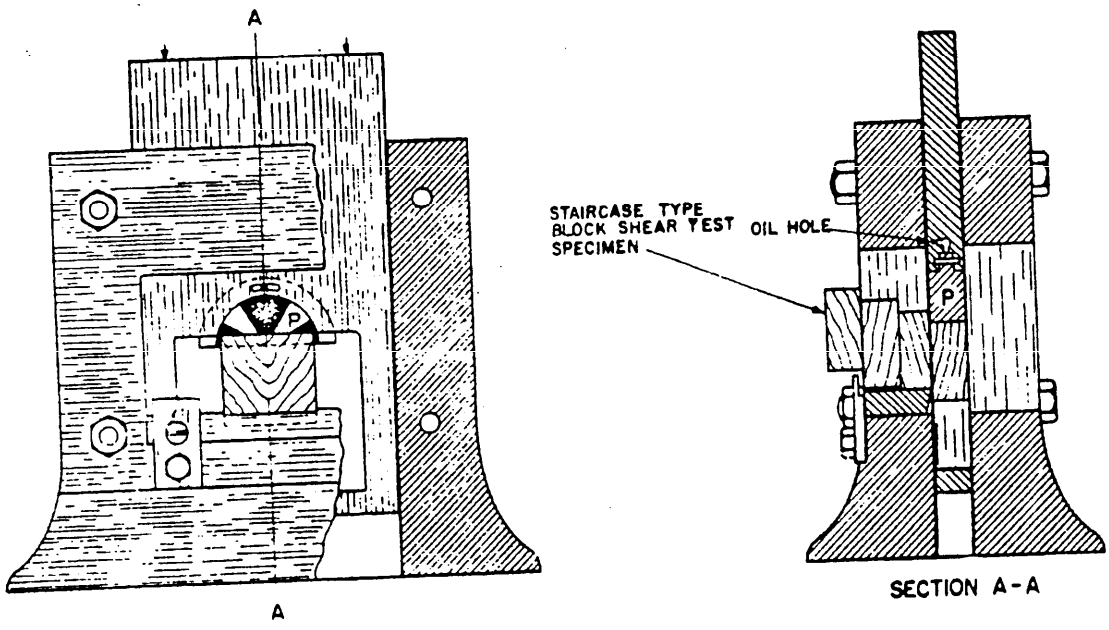


Figure 3 - Shearing Tool.

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APPENDIX - I

MINIMUM FACILITIES AND QUALITY CONTROL REQUIRED FOR PRODUCING WOOD LAMINATES FOR MARINE USE, AND RECOMMENDED MINIMUM PROCEDURES FOR DETERMINING THEIR ADEQUACY

10. SCOPE

10.1 This document discusses in detail the facilities which are considered necessary for the production of wood laminates for marine use, and recommends procedures for evaluating their adequacy for the use intended. Also, the requirement for quality control in successful marine laminating operation is delineated. In present Navy wood ship and boat construction wood laminates are replacing solid wood as structural components because laminates possess advantages of complete seasoning throughout, ability to be formed to shape in the process of assembly, prefabrication aspects, and greater strength. A laminated member can be built by assembling small pieces of wood without the use of fastenings. Solid wood of the proper grade and large dimensions is rapidly becoming depleted. A much greater supply of wood for ship and boat use thus becomes available through sawing low-grade material into boards, removing objectionable defects, and laminating. However, laminates for marine use require that the adhesive bonds must have unquestionable integrity. Achievement of this requisite requires use of special adhesives, extensive equipment, a high degree of technical skill, and maintenance of very close process control throughout production.

20. APPLICABLE DOCUMENTS

20.1 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FOREST PRODUCTS LABORATORY
Bulletin No. 1651 of October 1952 - Relative Humidity and Equilibrium Moisture Content Graphs and Tables for use in Kiln Drying Lumber.

(Copies may be obtained from the Director, Forest Products Laboratory, Madison 5, Wisconsin.)

AMERICAN SOCIETY FOR TESTING MATERIALS
ASTM Standard E4-61T - Verification of Testing Machines

(Copies may be obtained from American Society For Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

30. FACILITIES

30.1 Below is a descriptive list of facilities which are considered the minimum necessary to produce acceptable wood laminates for marine use. The quantity of each item is determined largely by the production schedule. However, the number and size of laminating jigs, curing chambers, and clamps required depend on the size, shape, and number of laminates to be produced. It is very important that all of the items be thoroughly checked for adequacy. Although many facility item which a plant has may be suitable for laminating more than one species of wood, other facility may be found suitable for laminating one species of wood only. Therefore, it is very important that facility item be checked for their applicability in fabricating laminates from the species of wood intended to be used in production.

30.2 Straight line rip saw.- For cutting lumber to the required width and removing defects. Any chain feed rip saw of good quality is satisfactory for this operation (not to be confused with 30.18.1 below which discusses a very special saw for preparing surfaces for subsequent edge gluing).

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30.3 Rough planer.- For surfacing rough lumber before cross cutting of defects, and before edge gluing and scarfing. Must provide true flat surfaces in order to give a proper reference bed for surfacing edges and scarfs for gluing.

30.4 Trim saw.- For trimming boards to the desired length and to remove defects.

30.5 Scarfing equipment.- For scarfing lumber to form laminations of the desired length. Must be capable of consistently producing the quality of scarf required in the laminating specifications.

30.5.1 Scarf cutter.- An acceptable method of cutting good scarfs is one which produces a good cut with no digging, tearing of the grain, chipping, or peeling between the annual rings. The scarf cutter must also produce a uniform cut across the board without splitting. To date the most satisfactory methods for consistently producing satisfactory scarf joints have been on a cabinet surfacer used with a form for holding the lumber, on a modified surfacer, on an Onsrude veneer scarfer, and on specially built machines. A modified radial arm saw with a bar type cutter head has also been used. However, it has generally not produced consistently satisfactory cuts, and is very difficult to keep in proper maintenance. If a surfacer is used for preparing scarfs, a preliminary cut to the approximate angle of the surface should be made by sawing or surfacing. The final scarf surface is prepared by removing a minimum cut. (If care is not taken, an initial cut by a saw will result in splitting of the feathered edge.) Forms for producing scarfs on a surfacer should be of sufficient length to assure uniform contact with the bed of the machine without danger of canting or clipping.

30.5.2 Clamps for scarf package.- The clamping assembly must be capable of safely applying and holding 150 - 200 p.s.i. pressure on all points of the glued assembly. The most reliable and economical method found to date for applying pressure to scarf joints during the curing process is in a package assembly. In this method a number of scarf assemblies are laid one above the other using a thin metal plate between each layer to prevent sticking together of layers, for heat conductivity, and for distribution of pressure. Using heavy oak cauls on top and bottom of the package, pressure is applied by means of calibrated rocker head clamps.

30.5.3 Curing equipment for scarfs.- Most satisfactory curing of scarfs is accomplished using the curing chamber for face laminating (see 30.9). However, other methods may be used providing temperature and humidity requirements of the laminating specification can be controlled adequately and consistently in production.

30.6 Cabinet surfacer.- For dressing laminations to the required thickness before assembling. The cabinet surfacer to be used for final surfacing of laminations must be capable of surfacing stock to a maximum tolerance of ± .005 inches. It is desirable that lumber be rough surfaced to approximate thickness before final surfacing and laminating. The maximum material to be removed by surfacing for laminating should not exceed 1/16 inch for each face. Knives should be carefully jointed to eliminate revolution marks. Planer speed should be so regulated as to insure a minimum of 20 and a maximum of 30 knife cuts per inch. It should be remembered that the cutting angle for hardwood should be less than for softwoods. The recommended cutting angle for oak should not exceed 15 degrees and for Southern pine and Douglas fir 20 degrees. A double head cabinet surfacer with ball bearing cutter heads has proved most satisfactory for dressing laminations. For limited production, a single head surfacer with ball bearings is satisfactory if tolerances for thickness can be maintained. If four-side matchers are used, particular attention must be given to the condition of the bed plates and pressure bars. It is very difficult to hold the necessary tolerances on molders or matchers.

30.7 Laminating jigs.- Must be capable of withstanding load under clamp-pressures and curing temperature conditions as outlined in laminating specifications. All laminating jigs must be of rigid construction. The type and size of laminating jig used will depend largely on type and size material to be laminated.

30.8 Laminate assembly clamps.- Application of correct, uniform pressure on a glue line is a prime requisite to the formation of a good bond. This requires that all clamps be uniformly spaced and uniformly tightened to the proper load which will deliver the required glue line pressure for the given clamp spacing and glue line area. For Southern pine and species of similar density the proper glue line pressure is 150

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± 25 p.s.i., and for oak is 175 ± 25 p.s.i. Clamps must be tightened again after 20-30 minutes of initial tightening. Inadequate attention to this feature has been one of the largest causes of failure in production. Yet, it is one of the easiest production items to control. To achieve this, clamps must be capable of applying and holding pressures required in the laminating specifications and withstanding repeated heating in the curing chamber without warping. The clamps must be equipped with rocker head type equalizing pressure heads to distribute pressure evenly on a laminated assembly. The pressure heads must be sturdily built and reinforced with webbing to prevent distortion and to permit even distribution of pressures across the whole member. In order that pressures along the length of a glue line can be uniformly applied and maintained, each individual clamp must have provisions for regulating the load applied, and easy means for determining this load. The most reliable clamps found to date are those which use a threaded shaft system to apply pressure. This type of clamp can be easily checked for adequacy through the use of a compressometer. Also, through the use of calibration data, the load delivered by a clamp can be easily determined through the use of a torque wrench. In selecting such clamps for use it is pointed out that acme threaded shafts, although generally a little more expensive, have less tendency to bind and hang under heavy load. They also will provide longer service. To date, clamping systems which use a hose containing water or compressed air to distribute pressure have not proven satisfactory.

30.9 Chamber for curing laminated assemblies.- The curing chamber used for curing laminates must be of such design as to provide proper uniform heat and humidity to the material. This requires that the walls of the chamber be insulated for both heat and vapor, and that the chamber be so constructed that air can circulate freely about the curing material, distributing uniform heat and humidity. Temperature and relative humidity conditions must be maintained uniformly throughout the process in the curing chamber. The purpose is to maintain the required temperature needed to cure the adhesive at all points of the glue line, and to keep the average moisture content of the wood essentially the same as it was before the period was begun. Also no degrade should result from the curing process. Several types of chambers have been successfully used, such as lumber dry kilns, specially built stationary curing chambers, or portable curing chambers.

30.10 Torque wrenches.- Should have at least two torque wrenches calibrated to the particular type of clamps used.

30.11 Adhesive.- The adhesive used for all marine laminating operations shall be an adhesive conforming to the requirements of MIL-A-22397.

30.11.1 Adhesive storage.- Must have adequate facilities for proper storage of adhesives in accordance with specifications.

30.11.2 Weighing mechanism.- Must have adequate weighing mechanism for weighing of adhesive components and for determination of glue line spread.

30.11.3 Equipment for mixing of adhesive.- This should be a mechanical type mixer capable of thoroughly mixing the adhesive to proper consistency without lumps, yet must be designed so that it will cause no bubbles or foam during the mixing process.

30.11.4 Equipment for spreading of adhesive.- Method of applying adhesive must be such that the correct amount of adhesive can be applied quickly. No foreign matter should be deposited on laminating stock which will interfere with the glue bond. For full scale laminating operations, commercial glue spreaders with rubber rolls, and which have provisions for adjusting rate of glue spread, have proven most successful. For smaller operations clean paint rollers which do not shed have also proven adequate.

30.12 Laminating room.- The laminating room should be such that it can be kept clean with a minimum of dust or particles present which will cause interferences with the glue bond. All wood working machinery must be equipped with dust collection systems. In addition, the room should be controlled so that room temperatures required in the laminating specifications can be met, yet to provide adequate ventilation for removal of objectionable adhesive solvent vapors which are given off.

30.13 Facilities for proper storage of lumber and dressed laminations.- Must have adequate facilities for storage of lumber under cover, including adequate space for storage of dressed laminations above 60°F.

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30.14 Equipment for handling lumber and laminates.- Adequate handling equipment must be provided for moving laminated members around the laminating shop and the yard without damage.

30.15 Shaping equipment.- Must have either a marine type band saw, or a conventional 36" band saw (heavy duty equipment), capable of carrying up to 1-1/2" blade. Filing room equipment adequate for servicing heavy duty band saws should be provided. If contour shaping to various bevels are contemplated, the band saw must be of tiltable type. In most operations a tiltable table will not work satisfactorily. In addition, if specifications or purchase orders require that laminates be dressed to specific dimensions after fabrication, suitable equipment for performing this operation is necessary.

30.16 Maintenance equipment.- Should have a grinding room adequately equipped for maintenance of cutter heads and saws. Also facilities should be available for maintenance of other equipment.

30.17 Testing equipment.-

30.17.1 Drying oven.- Must have a drying oven which is capable of maintaining a temperature of 105°C. (215°F.), to be used for determining moisture content and moisture distribution in lumber by the standard oven test.

30.17.2 Weighing apparatus.- Must have suitable mechanism for weighing lumber test samples to determine moisture content of wood. The weighing mechanism should be accurate enough so that weights can be correct within at least one percent and be able to indicate the weight of samples to the nearest 0.01 gram.

30.17.3 Moisture meter.- Must have an accurate moisture meter for quick determination of moisture content in lumber. Some moisture meters have a tendency to become inaccurate when batteries become depleted, or when room humidity conditions are adverse. Therefore, moisture meters should be checked for accuracy against standard oven tests at periodic intervals.

30.17.4 Potentiometer and thermocouples for accurate determination of glue line temperature in scarf-joints, laminate assemblies, and edge-joints.- Must be checked for accuracy through the use of a laboratory thermometer and a hot water bath. Calibration should include a full temperature range from 60°F. to 212°F. Although a potentiometer is generally used to measure glue line temperature it can also be used to determine ambient dry bulb and wet bulb chamber temperature conditions. This is done by using two units next to each other with a wet sleeve over one as is done with a normal wet and dry bulb system.

30.17.5 Compressometer.- Must have frequent periodic access to an accurate compressometer or equally effective portable apparatus for measuring compression loads for calibrating torque wrenches and clamps. The compression load measuring apparatus must be calibrated and certified for accuracy by a test laboratory.

30.17.6 Block shear testing apparatus.- Must have a suitable block shear testing apparatus to conduct block shear test. The shear test tool must be similar to that described in military specifications for wood laminates for marine use. The load measuring device shall be accurately calibrated and should include a damping apparatus which permits readings of ultimate loads under conditions of sudden failure. The loading device should give an accurate full scale reading of no more than 150 percent that of the maximum range of loads at which shear blocks will normally break. The accurately readable increments of load should not be greater than 1.5 percent of the maximum load reading on the scale and not greater than 5 percent of the minimum load at which the block shear test specimens will normally break. This is determined as follows: For example, if 8000 pounds is the anticipated maximum shear block breaking strength, the full scale reading on the dial gauge should not be greater than 8000 x 1.50, or 12,000 pounds. The maximum load reading on the dial is multiplied by 0.015. The minimum anticipated breaking strength is multiplied by 0.05. The lower of the two figures determines the minimum degree of accuracy required, and is therefore to be used as the maximum permissible dial increment figure. For example, if the maximum anticipated load is 8000 pounds, and the full scale reading on the dial is 12,000 pounds, the dial increment figure would be 12,000 x 0.015, or 180 pounds. If the minimum anticipated breaking strength were 3000 pounds, the dial increment figure would be 3000 x 0.05, or 150 pounds. Therefore, in order to test shear blocks which will normally break at loads ranging from 3000 pounds to 8000 pounds, with occasional higher loads not in excess of 12,000

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pounds, the scale on the dial must be accurately calibrated to increments of 150 pounds or less. It is pointed out that the increment of load determined in this manner is the maximum acceptable. For greater accuracy and ease in correlation of test results with an approved laboratory, it would be even more desirable for increment of load on the dial to be calibrated in smaller units. Another aspect which should be considered is the desirability of utilizing one apparatus for testing more than one species of wood laminates. In certain instances this can be accomplished by utilizing an accurately calibrated dial with smaller increments of load to provide a wider useful range. For example, if a dial with a maximum load reading of 12,000 pounds were calibrated in increments of 100 pounds, the apparatus could be utilized in testing shear block specimens which break at loads ranging from 100 - 0.05, or 2000 pounds to 8000 pounds, with occasional load not in excess of 12,000 pounds. An apparatus with such a dial should normally be able to be used in conducting the standard block shear test (standard shear area of 3 square inches) on most species of wood laminates fabricated for marine use.

30.17.7 Small band saw.- Small band saw for preparation of casehardening samples, moisture determination samples, and for preparation of block shear test specimens.

30.18 Miscellaneous laboratory equipment.- This includes such items as thermometers, gauges, micrometers, etc.

30.18.1 Edge-joining equipment (required only if edge-joints are to be fabricated).- Producing high quality edge-joints for use in marine laminates requires adherence to very close tolerances in equipment and methods. The extent of marine laminating, the type of marine laminates being produced, and the availability of equipment and experienced personnel will determine the practicability of using edge-joined material. It will also determine whether the laminator should do his own edge-joining or whether he will find it more advantageous to purchase edge-joined material from a manufacturer who has already established his qualifications as a supplier of edge-joined dimension stock. Below is a list of additional equipment which would be necessary for a laminator to produce his own edge joints, and procedures for checking the adequacy of this equipment.

30.18.2 Equipment for preparing surfaces for edge-joining.- Equipment for producing surfaces for edge-joining should be of reasonably late design and must be in first class condition. The equipment must be capable of preparing edge-joint surfaces on the longest length of edge-glued material expected to be used in production. If a chain feed straight line edge-joining rip saw is used for this purpose, it requires that the ball bearings on the saw arbor, and the grooves and track on the chain be in first class condition. The saw must be well balanced and fitted. For limited production, surfacing of edge joints has been satisfactorily accomplished on a molder that is kept in the very best of condition. For very limited production, surfacing of edge joints can be successfully accomplished on a hand jointer which is maintained by an expert millwright. However, this last method is a very cumbersome and expensive procedure and is probably practical only for short (4' or less) length material. Whichever method is used, the resulting edge joints should be carefully checked for smoothness and evenness of cut. Assurance should also be made that the planes of the edge joint surfaces are perpendicular to the planes of the flat surfaces of the lamination, and that the prepared surfaces will match when assembled. Special care must be given in matching edge-joint surfaces prepared on a jointer.

30.18.3 Clamps for edge-joining.- Application of the correct, uniform pressure on a glue line is a prime requisite to the formation of a good bond. This requires that all clamps be uniformly spaced and uniformly tightened to the proper load which will deliver the required glue line pressure for the given clamp spacing and glue line area. For Southern pine and species of similar density the proper glue line pressure is 150 ± 25 p.s.i., and for oak is 175 ± 25 p.s.i. If pressure must be maintained for a considerable length of time, clamps should be retightened after 20-30 minutes of initial tightening. Inadequate attention to this feature has been one of the largest causes of failure in production; yet, it is one of the easiest production items to control, particularly if a standard clamp spacing, or a few standard clamp spacings are used for all edge-gluing operations. In edge-gluing lumber for width, the maximum clamp spacing measured on centers should not exceed 14 inches. Closer clamp spacing is often necessary to more uniformly distribute the load supplied by the clamps to the glue line, and to assure that the clamp loading heads do not crush the wood. The most satisfactory methods found to date for applying pressure to edge-joint glue lines are stock type clamp carriers and stacking clamps. Stacking clamps have generally been found much more adaptable

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to producing edge joints for laminations for marine use. This is due to their low initial cost, versatility in clamping various lengths of material (particularly lumber over 8 feet in length), ease of handling and storage, low maintenance costs, and ability to hold material flat. They also generally require less extensive curing facilities. Both stacking clamps and clamp carriers have threaded shafts for applying pressure to each individual clamp head, which can be readily calibrated to determine the load delivered for a given torque. To date, other clamping systems such as hydraulic or air clamps have not proven satisfactory. This is primarily due to the difficulty of determining and regulating pressures along the full length of the glue line. Therefore, unless these features can be provided, the use of fluid or air pressure is not considered acceptable.

30.18.4 Curing edge joints.- (see 30.5.3).

40. QUALITY CONTROL REQUIREMENTS

40.1 In addition to adequate equipment, a successful marine laminating operation requires adequately trained personnel, and an effective quality control procedure.

40.2 Supervisory technical personnel.- Supervisory technical personnel require a high degree of technical knowledge in wood working operations and equipment, wood technology, and an ability to understand and supervise marine laminating operations within the scope of governing specification requirements. Previous experience in commercial structural laminating operations is usually helpful. However, it is pointed out that in general marine laminating operations are much more exacting than most commercial operations. Therefore, a sound foundation is necessary regarding these differences in order to avoid failures in production.

40.3 Production personnel.- In order to make a marine laminating operation a success, personnel must be thoroughly indoctrinated in the materials and process requirements of the applicable military specifications.

40.4 Maintenance personnel.- Because of the consistently reliable performance to close tolerances required of equipment for marine laminating operations, a firm engaged in such operations must have first class maintenance personnel to service this equipment.

40.5 Quality control engineer and inspection personnel.- Marine laminating requires close quality control and inspection procedures throughout the process to maintain the required close tolerances in production processes. A trained quality control engineer is required to analyze specification requirements and to devise definite procedures by which these requirements can be checked and documented. Trained inspectors, under the supervision of the quality control engineer, are required to inspect for conformity with the process and material requirements of the specification by the procedures developed by the quality control engineer. Trained personnel are also required to conduct such laboratory tests as block shear tests, make moisture content and casehardening determinations, and to check accuracy of instruments, devices such as controls, moisture meters, torque wrenches, etc. To be most efficient, the quality control organization should be independent of the production department.

50. CALIBRATION OF VARIOUS EQUIPMENTS

50.1 Certain equipments used for marine laminating operations require calibration to assure correct performance. Below are recommended procedures for calibrating such equipments. Whether or not a piece of equipment remains in calibration depends on use and quality. During production recalibration must be such frequency as to provide accuracy with reasonable assurance at all times.

50.2 Calibration of clamps.- Assuming that clamps with threaded shaft systems are used for applying glue line pressure, the following procedure is used for their calibration with an accurate compressometer and an accurate torque wrench.

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50.2.1 Calibration procedure.- Insert the compressometer between the pressure heads of the clamp. It may be necessary to insert oak cauls between the pressure head of the clamp and the bearing surfaces of the compressometer. Using the torque wrench, tighten the clamps to compress the compressometer. The applied load can be read on the dial gauge of the compressometer in pounds, and the corresponding torque on the torque wrench can be read in ft.-lbs. A full range of load readings should be taken. As in production use, all clamps should be well greased and be in maintenance. This will aid in preventing hanging or binding due to an abnormally high co-efficient of friction on the screw. Care should also be taken to assure that the compressometer dial actually reads total load in pounds. If the dial reads fluid pressure in pounds per square inch, this reading must be converted to total load readings in pounds. By plotting on graph paper observed torque readings against the corresponding load readings, the safe capacity for the clamp can be determined. As noted in figures 1A and 2A, a straight line can usually be drawn through the first group of points. Then at a given load the points deviate from a straight line. The point on the graph where the curve deviates from the straight line relationship is known as the yield point or elastic limit and is the point at which weakening of the metal takes place. Clamps should not be loaded beyond the yield point. A suitable number of clamps of each type used should be calibrated to assure adequate representation.

50.2.2 Determination of proper clamp spacing and proper load applied by clamps. The proper load to be applied by the clamps for any given uniform clamp spacing and lamination width can be readily determined from the calibration information as shown by the following:

50.2.2.1 Clamps for clamping final laminate assemblies.- Assume that clamps are spaced on 7 inch centers and that the material to be bonded is 7 inches wide. For oak the required glue line pressure is 175 ± 25 p.s.i. Assume for practical purposes that the load applied by each uniformly spaced clamp is uniformly distributed for a distance of at least 3.5 inches to each side of the clamp, or over an area of 49 (7x7) square inches. (This assumption can be made only if clamps are spaced close enough together, and the cauls and/or the material are rigid enough to uniformly distribute the load.) In order to apply the required 175 p.s.i. pressure to the glue lines, each individual clamp must supply a load of 175 p.s.i. x 49 square inches, or approximately 8575 pounds. Similarly computed for a minimum glue line pressure of 150 p.s.i., each clamp would be required to exert a load of 7350 pounds. At the maximum 200 p.s.i. pressure range, each clamp would be required to supply a load of 9800 pounds. As noted in the example calibration graph, (Figure 1A) the particular clamps could be safely loaded beyond 9800 pounds without damage. Also taken from the graph, a torque reading of 93 ft.-lbs. on the torque wrench would be required for the clamp to exert a load of 7350 pounds. 109 ft.-lbs. would correspond to a load of 8575 pounds. 127 ft.-lbs. would correspond to a load of 9800 pounds. Therefore, using these particular clamps and the above cited clamp spacing and lamination width, a torque reading of 109 ± 16 ft.-lbs. would be required to maintain a glue line pressure of 175 ± 25 p.s.i. If from the calibration graph it is found that the clamps will not safely take a load of at least 9800 pounds without deforming beyond the elastic limit, heavier clamps are necessary for use with that width material and clamp spacing. Closer spacing of clamps will, of course, require less load to be supplied by each clamp. For example, if the particular clamps described above were spaced on 5-inch centers, the load required to be delivered by each clamp to supply a pressure of 200 p.s.i. at the glue line would be 7000 pounds (5" x 7" x 200 p.s.i.). This is 3800 pounds less than the load required to be delivered if clamps were spaced on 7-inch centers.

50.2.2.2 Clamps for edge-gluing.- Assume that clamps are spaced on centers at the maximum recommended distance of 14 inches, and that the material to be edge glued is 7/8 inch in thickness. For oak, the required glue line pressure is 175 ± 25 p.s.i. Assume for practical purposes that the load applied by each uniformly spaced clamp is uniformly distributed for a distance of at least 7 inches to either side of the clamp, or over an area of 12.25 (7/8 x 14) square inches. (This assumption can be made only if clamps are spaced close enough together, and the cauls and/or the material are rigid enough to uniformly distribute the load.) In order to apply the required 175 p.s.i. pressure to the edge glue line of oak, each individual clamp must supply a load of 175 p.s.i. x 12.25 square inches, or approximately 2144 pounds. Similarly computed for a minimum pressure of 150 p.s.i. and a maximum of 200 p.s.i. the range allowed would be not less than 1838 pounds nor more than 2450 pounds respectively. The torque necessary to tighten the clamp to these respective loads can be taken directly from the graph plotted from the results of calibration of clamps with the compressometer. From the example graph shown in Figure 2A, the torques required to be applied by the torque wrench to supply 1838, 2144, and 2450 pound loads respectively would be 35, 38.8, and 42.6 ft.-lbs. Rounded off these would be approximately 35, 39, and 43 ft.-lbs. Therefore, using these particular clamps

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and the 7 8-inch lumber thickness and 14-inch clamp spacing cited above, a torque reading of 39 ± 4 ft.-lbs. would be required to maintain a glue line pressure of 175 ± 25 p.s.i.

50.2.2.2.1 Production use of clamps.- In the production use of clamps, care should be taken to assure that the loading heads on the clamps do not damage the wood, particularly directly under the loading heads. If damage does take place, the load supplied by each clamp is too great. This can be easily corrected by reducing the clamp load by a suitable amount, and recalculating a shorter uniform clamp spacing which will provide the required glue line pressure.

50.3 Calibration of torque wrenches.- Torque wrenches are calibrated in conjunction with clamps through use of an accurately calibrated compressometer as outlined above.

50.4 Compressometer.- The compressometer or equally effective portable device for measuring compression loads should be calibrated by comparing various loads indicated by it with those of a standard test machine, which in turn has been recently verified for accuracy, by one of the methods listed in (ASTM) standard E4-61T "Verification of Testing Machines." Loads should be calibrated at frequent intervals throughout the full range of the load indicating dial of the compressometer. For visual comparative purposes, a graph should be drawn in which corresponding loads shown by the compressometer and the standard test machine are plotted against each other.

50.5 Calibration of temperature and humidity indicating and recording instruments used in curing laminate assemblies, scarf-joints, and edge-joints.- A complete calibration should be made of all temperature and humidity recording and indicating instruments by means of a hot water bath and an accurate laboratory thermometer. This calibration should also include potentiometer and thermocouple units. The calibration range should run from approximately 60°F. to approximately 212°F. Calibration of indicating and recording temperature units are accomplished as follows:

50.5.1 Calibration procedure.- Clean all temperature elements thoroughly and immerse in a large glass container of distilled water at a temperature of approximately 60°F. Also in the water bath insert an accurate laboratory thermometer which is graduated in increments of not larger than 1°F. Stir the water around the temperature actuating units until the indicating or recording instruments show an equilibrium with the temperature of the water. Record the temperature indicated on the laboratory thermometer and corresponding readings indicated or recorded by the other instruments. Raise the temperature of the water in the container an increment of approximately 15°F. Stir the water in the container until the indicating or recording temperature actuating units are in equilibrium with the temperature of the water. Again record the temperature indicated on the laboratory thermometer and the corresponding readings taken from the instruments being calibrated. In a like manner, the calibration should be extended to a useful range of approximately 212°F.

50.6 Weighing mechanisms.- Should be calibrated for accuracy by use of standard weights.

50.7 Moisture meters.- A simple procedure for checking accuracy of a moisture meter is to take readings with the moisture meter on a few typical pieces of lumber. Within a short time thereafter, cut sections out of the boards where the moisture readings were taken; weigh the small sections with the weighing mechanism described in 30.11.2, and dry in an oven kept at 105°C. (215°F.) until there is no appreciable loss in weight (for 1-1/2" x 1-1/2" square block of 1" nominal material drying takes approximately 6 to 8 hours).

$$\text{Percent moisture content (based on oven dry weight)} = \frac{\text{wt. before drying - wt. after drying}}{\text{wt. after drying}} \times 100$$

50.8 Block shear testing apparatus.- Any of the methods listed in American Society For Testing Materials Designation E4-61T "Verification of Testing Machines" is considered acceptable for calibration or verification of the shear test machine. Another permissible alternate would be to compare various loads registered by the laminator's apparatus with those of a standard test machine which has been recently verified for accuracy. A full range of loads indicated by the dial of the test apparatus being calibrated should be

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verified. A graph should be drawn in which corresponding values of the block shear test apparatus and the verification apparatus are plotted against each other.

60. PERFORMANCE TYPE TESTS FOR DETERMINING ADEQUACY OF VARIOUS EQUIPMENT

60.1 During production determining performance of equipments must be of such frequency as to provide required performance with reasonable assurance at all times.

60.2 Methods for Testing Scarf Joints.- The cyclical exposure test as outlined in applicable military wood laminate specifications is considered a suitable method for evaluating adequacy of glue line in scarf-joints (provided all process requirements have been met). For daily checks of scarfs prior to face laminating the methods of CS-253 are applicable.

60.3 Methods of testing edge joints.- The cyclical exposure test as specified in applicable military wood laminate specifications is considered a suitable method for evaluating adequacy of edge joint glue lines (provided all process requirements have been met). For a daily check, the block shear test is recommended to give information prior to the exposure test.

60.4 Testing of edge joints by the block shear test method.- For preliminary testing of edge joints full length boards containing at least two edge joints should be produced on regular production equipment in accordance with specifications. The boards prepared should be of at least two thicknesses, 1/2" and of a large thickness, and at least 7 boards of each thickness should be prepared. At least seven samples should be taken at random from each edge joint of each edge glued board and tested in block shear. The block shear strength must be not less than that specified in the particular laminating specification for the species of wood listed for block shear strength between laminations. The accompanying percent wood failure shall be not less than 75 percent. These tests should be conducted at a laboratory by qualified personnel on standard test equipment and in accordance with standard test procedures except for width of test specimen.

60.5 Testing proficiency of cabinet surfacer.- Twelve boards of the species intended to be laminated in production should be used for checking machining performance. The boards should be nominal 1-inch, approximately 6 inches wide, and at least 6 feet long. At a given thickness setting, one board should be run through the center of the surfacer, one through the left side, and one through the right side. This procedure should be followed four times. Each board should be thoroughly identified. All boards should then be measured for tolerances with a micrometer, accurate to the nearest 0.001 inch. Measurements should be taken at the center of the length of each board and at both ends. Two measurements should be taken at each of the three locations on the board to measure variation in thickness across the width of a given board. By this procedure machining variation in thickness can be determined along the length of a board, across its width and at various planer bed locations. This data should be presented in a table similar to Figure 3A. Throughout the length or width of a board no two measurements taken should exceed the maximum tolerance permitted by the applicable laminate specification. In addition, surfaces should show no manufacturing defects such as skip, torn grain, raised grain, chipped grain, and chip marks, to an extent which will prevent sufficient contact between gluing surfaces for complete adhesion.

60.6 Testing distribution and adequacy of temperature and humidity conditions in a curing chamber.-

60.6.1 Test procedures and results.-

60.6.1.1 Place at least three laminated assemblies at various locations in the curing chamber, inserting thermocouples at the innermost glue line of each assembly. This should preferably be done when the chamber is full.

60.6.1.2 Insert at least three temperature and humidity recording or indicating devices in various parts of the curing chamber.

60.6.1.3 Make a drawing of the setup showing distribution of samples and instruments.

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60.6.1.4 Submit samples and chamber to a full curing cycle, being careful that in the heating and cooling periods the ambient chamber temperature is not increased or decreased by a rate faster than that permitted by the governing specification. During the required curing period, no glue line temperature in any sample should be less than that required for the approved adhesive and wood species laminated.

60.6.1.5 Throughout the curing process relative humidity conditions should be uniformly maintained throughout the curing chamber which will keep the average moisture content of the wood essentially the same as it was before starting the curing process. The samples should also show no signs of warp, case-hardening, checking, or other degrade as a result of the curing process.

60.6.2 Maintaining suitable relative humidity conditions in curing chamber.- A number of charts and graphs are available which show the correct relative humidity conditions which must be maintained in a curing chamber for a given chamber temperature condition to hold wood at a given moisture content. Unfortunately, many of these tables or graphs have been misinterpreted. A good source of practical, understandable information on the subject is Forest Products Laboratory Bulletin No. 1651 of October 1952 entitled, "Relative Humidity and Equilibrium Moisture Content Graphs and Tables for use in Kiln Drying Lumber." Table I of the referenced bulletin shows relative humidity conditions which must be maintained at a given temperature condition to reach a given equilibrium moisture content in wood. As noted in the table, different temperature conditions generally require different relative humidities in a chamber to maintain the same equilibrium moisture content condition in wood. Therefore, during the varying temperature cycle followed in curing laminate assemblies, it is also necessary to vary relative humidity conditions in the curing chamber, if the wood is to remain at essentially the same moisture content throughout the curing process.

60.6.2.1 Difficulty in maintaining enough relative humidity in the curing chamber.- At times laminators have had difficulty in maintaining high enough relative humidity conditions in the curing chamber. This problem usually originates when steam humidification is used. A common error in the use of steam humidification has been the straight injection of steam from steam heating system into the curing chamber. The line pressure of the steam is usually 10 to 30 p.s.i. Difficulty can be avoided by either using fine water sprays or installing a system in the steam spray line which will completely saturate the steam as it leaves the pressure system.

60.6.2.2 Condensation in the curing chamber.- In maintaining proper relative humidity conditions in the curing chamber during the curing cycle, some firms have experienced difficulty with heavy condensation. Heavy condensation can usually be attributed to one or more of the following:

60.6.2.2.1 Trying to lay up laminated assemblies at too low temperatures.

60.6.2.2.2 Inadequate insulation and vapor barrier materials in the curing chamber walls.

60.6.2.2.3 Insufficient air circulation in the chamber.

60.6.2.2.4 Increasing the temperature in the curing chamber at too rapid a rate. Heating a chamber slowly will avoid condensation particularly at the higher temperatures where a greater actual amount of water is present for a given relative humidity condition. For example, if air at a temperature of 80°F. and 67 percent relative humidity approximately 7.8 grains of water per cubic foot was cooled to 60°F. (by chamber walls, clamps, etc.) approximately 2 grains of water would condense per cubic foot of air. This is a relatively small amount and could be eliminated by using slightly less than equilibrium moisture content conditions in the curing chamber during initial part of the heating period. However, if the air temperature was maintained at 160°F. and 79 percent relative humidity and if the chamber wall was at 140°F. temperature approximately 14 grains of water would condense per cubic foot of air. This is about 7 times as much condensation as would occur with the same 20°F. temperature spread at the lower temperature range. It is therefore obvious that if heavy condensation is to be avoided at the higher temperatures, better circulation, more adequate chamber insulation, and slower heating conditions are necessary. A graph showing these temperature-humidity relationships is shown in Figure 4 of the above referenced Forest Products Laboratory Bulletin No. 1651.

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70. PROFICIENCY IN CONDUCTING BLOCK SHEAR AND CYCLICAL DELAMINATION TESTS

70.1 Correlation tests.- Correlation tests with a laboratory approved by the Bureau are required to show that results of block shear and cyclical delamination tests conducted at the laminator's plant are essentially the same. The correlation tests should be conducted in a manner similar to that described below using the species of wood intended to be laminated in production:

70.1.1 Preparation of test specimens.- To effectively correlate tests conducted at two testing stations, samples must be matched. Matching is accomplished by cutting full cross sections from several laminates. Enough sections should be cut to provide at least five sections for block shear tests and five for cyclical delamination tests. Each section should contain at least five gluelines.

70.1.1.1 Block shear tests.- For block shear tests the laminate sections should be long enough and wide enough so that each can be cut into four quarters (see figure 4); each quarter long enough that a stair-step specimen representing all gluelines can be easily prepared from it. Each laminate section should be carefully identified as A, B, C, D, etc. The four quarter sections in each laminate section shall be identified as I, II, III AND IV. Quarter I shall be diagonal to quarter III, and quarter II diagonal to quarter IV. On each quarter individual gluelines shall be marked 1, 2, 3, and 4, etc. After marking the laminate section as described above, it shall be cut into quarters.

70.1.1.2 Cyclical delamination tests.- Sections cut from laminates for delamination tests should be marked for two adjacent specimens each 4 inches long and of full cross section. Identify each laminate section as A, B, C, etc. After marking, cut the two specimens from each laminate section.

70.1.2 Wrapping and shipping of specimens.- Each block shear quarter or delamination specimen should be sealed in polyethlene bags or other effective moisture barrier material. The preparing activity should retain block shear quarters I and III and delamination specimens I from each laminate section. Quarters II and IV and specimens II should be forwarded to the approved laboratory.

70.1.3 Cutting test specimens.- On a date mutually acceptable, both activities should cut test specimens and conduct tests.

70.1.4 Test procedures.- Block shear and delamination test specimens should be prepared and tested in accordance with the applicable laminate specifications. Block shear specimens shall be evaluated for shear strength and wood failure.

70.1.5 Results and analysis.- The results of the tests shall be tabulated to show the shear strength (in psi), wood failure (percent) and delamination (percent) for each glueline of each block shear quarter and delamination specimen. The average of the gluelines in each quarter and delamination specimen shall also be shown in the table of results. Analysis of the results should be by statistical methods to show the variation of the preparing activity from the approved laboratory. The analysis should be performed on two sets of data to show extent of correlation of results.

(1) Individual gluelines.
(2) Average for each laminate section.

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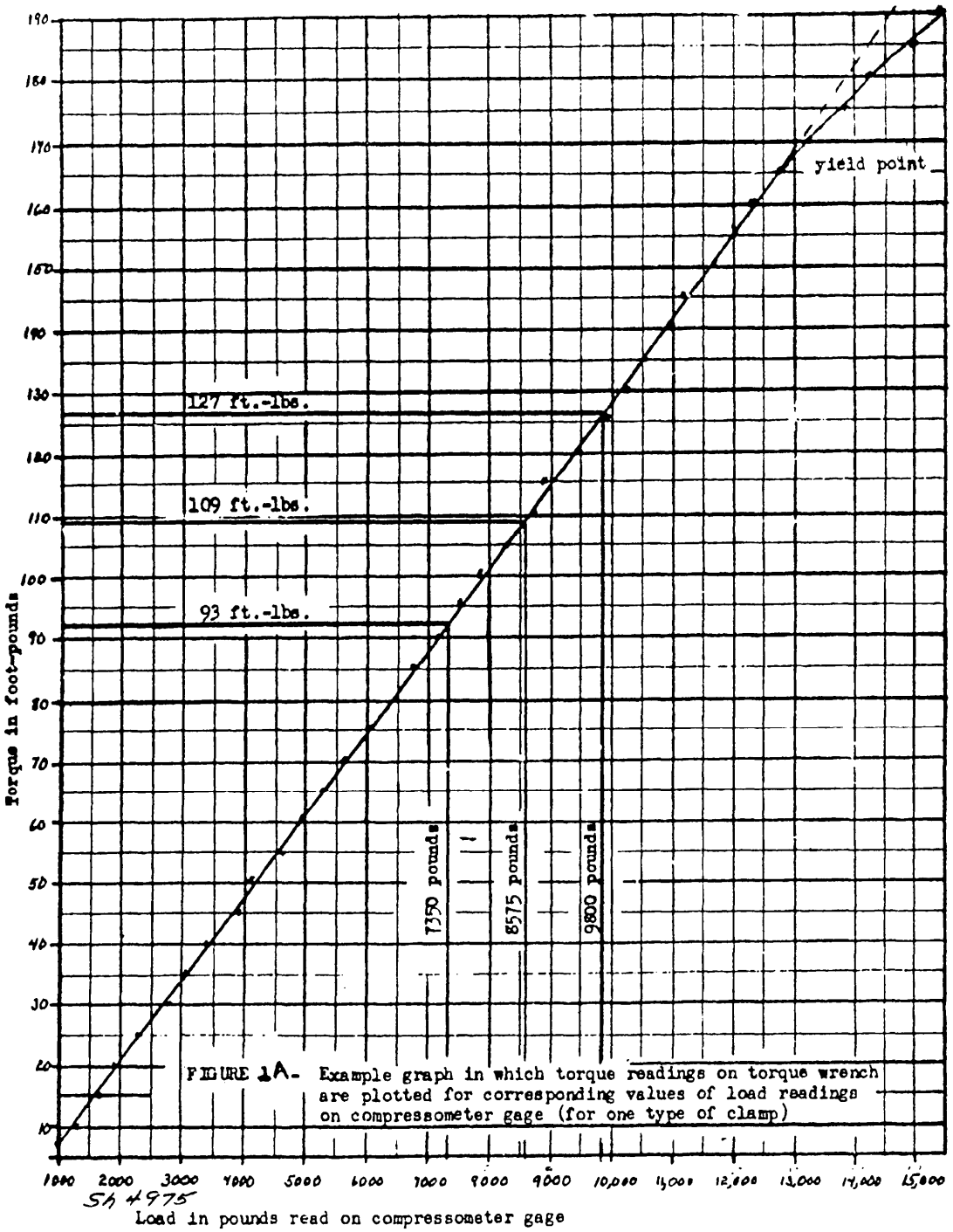


FIGURE 1A. Example graph in which torque readings on torque wrench are plotted for corresponding values of load readings on compressometer gage (for one type of clamp)

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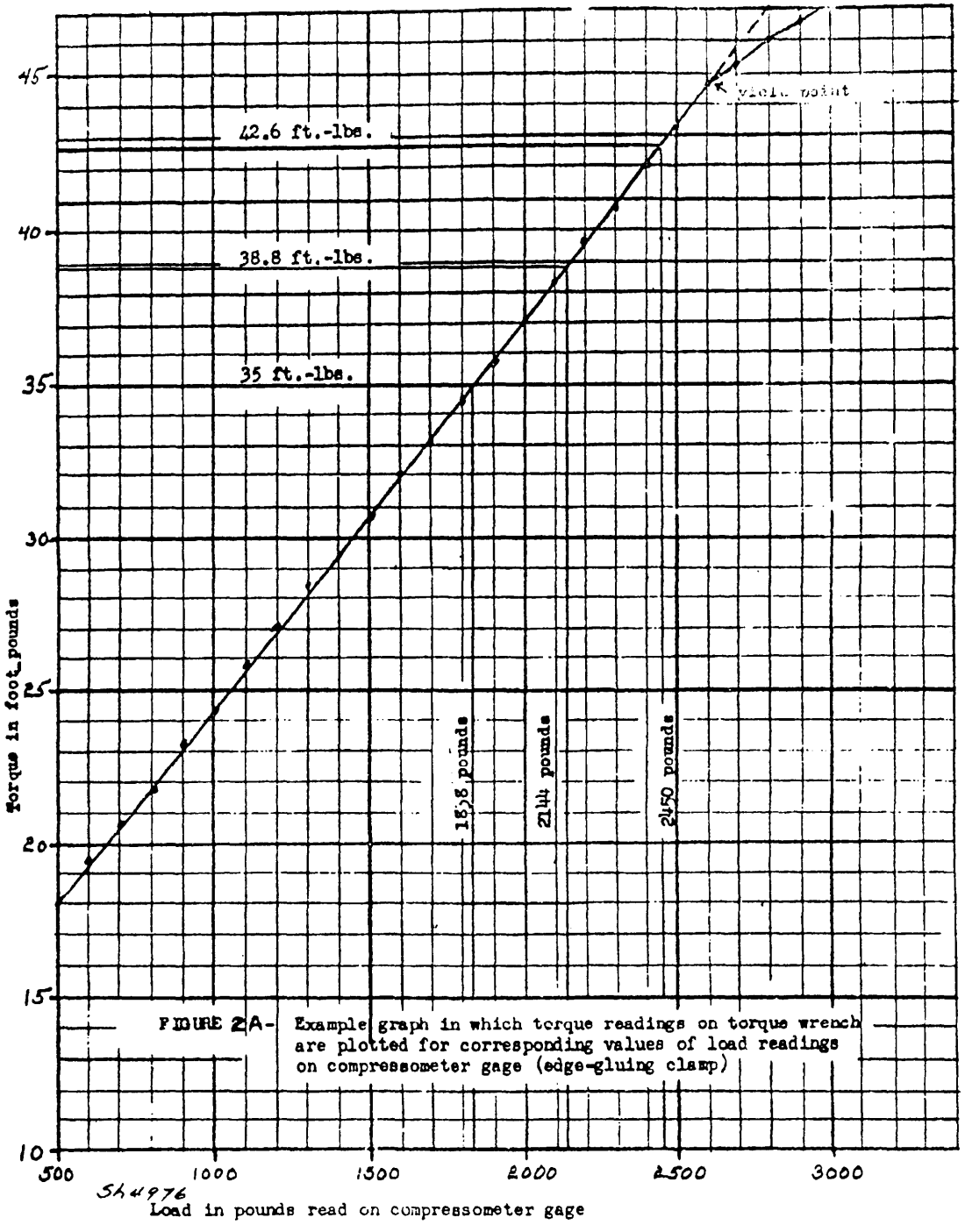


FIGURE 2A- Example graph in which torque readings on torque wrench are plotted for corresponding values of load readings on compressometer gage (edge-gluing clamp)

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Boards through left side of surfacer (inch)						Boards through center of surfacer (inch)						Boards through right side of surfacer (inch)					
Leading End		Center		Aft End		Leading End		Center		Aft End		Leading End		Center		Aft End	
Left Side	Right Side	Left Side	Right Side	Left Side	Right Side	Left Side	Right Side	Left Side	Right Side	Left Side	Right Side	Left Side	Right Side	Left Side	Right Side	Left Side	Right Side
0.750	0.751	0.749	0.751	0.750	0.752	0.752	0.751	0.749	0.748	0.748	0.750	0.746	0.748	0.747	0.749	0.750	0.748
0.757	0.752	0.754	0.753	0.751	0.750	0.753	0.752	0.748	0.746	0.748	0.749	0.749	0.750	0.748	0.749	0.750	0.749
0.753	0.752	0.751	0.754	0.751	0.752	0.752	0.753	0.750	0.747	0.749	0.750	0.747	0.749	0.749	0.750	0.747	0.750
0.751	0.753	0.752	0.750	0.752	0.751	0.751	0.752	0.750	0.751	0.750	0.751	0.749	0.748	0.750	0.749	0.750	0.748

Figure 3 - Sample chart showing thickness tolerances of boards run through a surfacer.

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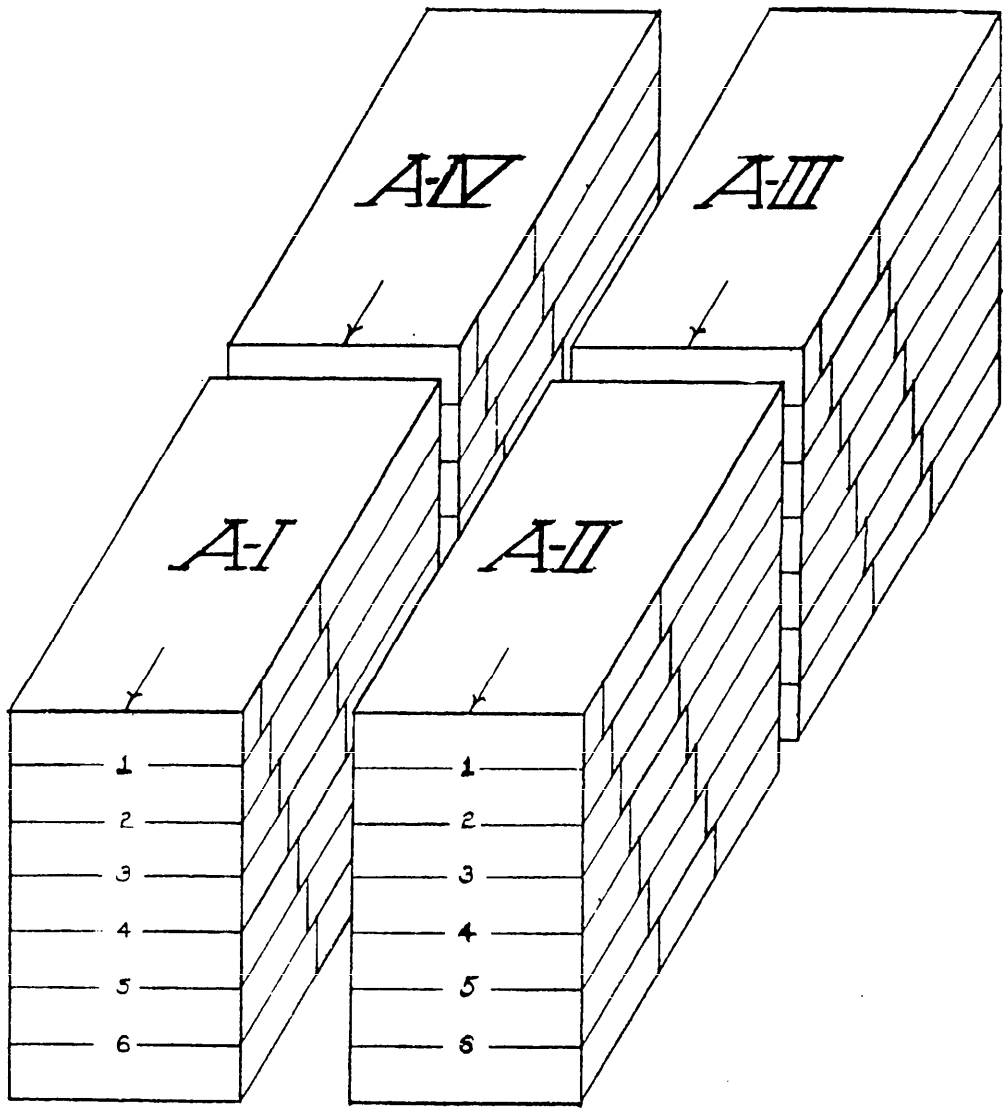


Figure 4 - Example of a Wood Laminate Block Shear Test Correlation Sample.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4963, WHICH IS OBSOLETE

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